

NinjaHeads: Gaze-Oriented Parallel View System for Asynchronous Tasks

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ABSTRACT

Virtual parallel experience enables a person in Virtual Reality (VR) to receive multiple and concurrent perspectives. It also enables the person to synchronously participate in activities or solve multiple tasks at once. This research focuses on utilizing parallel views for asynchronous tasks by providing the user with intuitively extended view perspectives driven by gaze. In this demonstration, we present NinjaHeads, a parallel view system that generates four additional perspectives "parallel views", based on the user's gaze point. We introduce a technique where a user can manipulate parallel views and their behaviors through eye, head, and hand movements. This allows the user to concurrently examine an object or a point of interest from different perspectives, as if the user has more than a pair of eyes.

CCS CONCEPTS

• Human-centered computing → Information visualization.

KEYWORDS

parallel view, virtual reality, human augmentation

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1 INTRODUCTION

Virtual Reality (VR) Head Mounted Display (HMD) enables users to immerse themselves in virtual environments and explore digital worlds. Over time, novel techniques have been proposed to support virtual exploration and navigation activities. One of these approaches, known as the parallel experience [Kasahara et al. 2016], allows users to simultaneously immerse themselves in multiple

virtual environments, obtaining multiple perspectives. This makes it possible to participate in synchronous activities such as playing parallel ping-pongs [Takada et al. 2022]. However, there are limited number of prior works that focused on utilizing the parallel experience for asynchronous activities, despite its potential to support exploration and interaction tasks [Teo et al. 2023].

We introduce NinjaHeads, a prototype system that utilizes parallel views to support virtual asynchronous tasks by providing users with additional perspectives based on their gaze point. When the user gazes at a location, they can explore around and behind the point of interest through four additional perspectives. Furthermore, the user can use specific hand gestures to manipulate the view distance and behavior. Together, this allows users to observe an object from various viewing angles, improving task precision and depth perception, even in the presence of blind spots or view occlusions. This paper discusses the design and development of NinjaHeads-driven parallel view and its applications in explorational and navigational activities. It presents concepts for using a parallel view system to support tasks involving the assessment of objects and spaces, such as data analytics, architecture, and scene investigation.

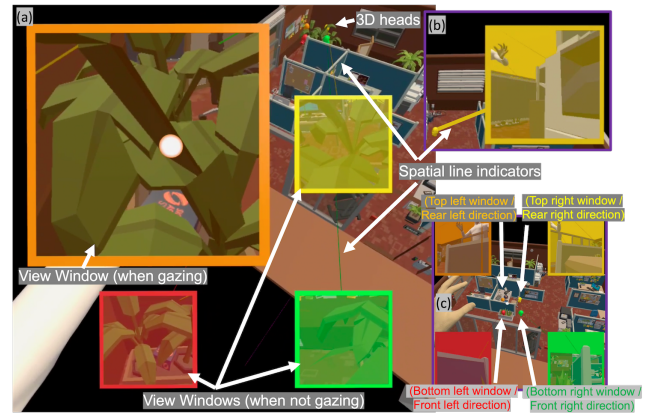


Figure 1: The NinjaHeads system overview. A view window expands and becomes clearer when gazed upon (a), and returns to the original size and opacity otherwise (c). Spatial line indicators are employed to provide visual guidance on the origin point of view (b).

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2 SYSTEM DESIGN

The system operates on a Meta Quest Pro VR HMD. In the VR scene, four floating heads called 'NinjaHeads' indicate parallel views that are rendered synchronously and around the point (or object) of interest following the user's eye gaze. Each NinjaHead represents an additional perspective and consists of a view window, a colored 3D head, and a spatial line indicator. The NinjaHeads are arranged in an 'X' shape pattern, with the intersection point placed at the user's gaze point and a NinjaHead positioned at each end looking toward the intersection point from 10cm away. The system utilizes sensors on the VR HMD to track the user's head, eye, and hand movements. These data are processed differently but serve a common purpose: enabling the user to navigate and operate the NinjaHeads in the VR environment.

2.1 NinjaHeads behaviour and implementation

The system tracks and stores the user's gaze point every 0.12 seconds (every 8 frames). It compares the current and previous gaze points and updates the position of the NinjaHeads based on the Euclidean distance between the two points. When the distance is $>15\text{cm}$, the NinjaHeads remain at their current positions while updating the previous gaze point. However, when the distance is $<15\text{cm}$, both the NinjaHeads' positions and the previous gaze point are updated with the current gaze point. Notably, the gaze point does not update when the user is gazing at the view window. This prevents the NinjaHeads from moving towards the view window while the user is examining a point of interest through the parallel views. This design choice allows the user to intuitively control the NinjaHeads and avoids unwanted gaze updates when the user's gaze shifts from a point of interest to the view window.

The view windows presents the visual outputs from the NinjaHeads. The system arranges the view windows based on their relative positions to the colored 3D heads and presents them in front of the user's main perspective (see Fig.1, (C)). To reduce visual clutter, the view windows have an original transparency of 75%. However, when the user gazes at a view window (see Fig.1, (A)), its transparency increases to 95% and its size increases by 1.5x. This allows the user to examine the content clearly and easily. Additionally, there is a line connecting each 3D head and view panel, using respective colors to indicate the origin point of view (see Fig.1, (B))

2.2 Hand Gestures for View Control

The NinjaHeads can be controlled by making a pinch hand gesture. When the user performs this gesture, the system stops moving the NinjaHeads based on gaze. Instead, the user can rotate their head, causing the NinjaHeads to rotate symmetrically in the same direction. This enables the user to explore a broader view of the content in the parallel views. Additionally, the user can make two-hand pinch gesture (double pinch) to activate the distance adjustment mode. In this mode, the user can adjust the distance of the NinjaHeads from the gaze point (see Fig.2) using hand movement. The distance can be increased up to 100cm away from the gaze point when the hands are farther, or decreased to no closer than 0.1cm when the hands are closer.



Figure 2: Using pinch hand gestures to adjust the view distance between the NinjaHeads and the gaze point.

3 USER EXPERIENCE

The concept of NinjaHeads offers significant benefits in various applications. In the field of architecture and design, NinjaHeads can support the exploration and assessment of virtual models of buildings, helping architects and designers to make more informed decisions regarding design elements, spatial relationships, and lighting conditions. At SIGGRAPH Asia 2023, we will showcase an example of a building inspection, allowing attendees to use NinjaHeads to intuitively examine every detail of a virtual structure.

4 CONCLUSION

We present NinjaHeads, a system that utilizes parallel views to provide users with the ability to examine various perspectives of a point of interest in VR. The system generates five (one main perspective and four additional) perspectives, around a specific point or object of interest. It renders these additional perspectives based on the user's gaze point and utilizes hand gestures to adjust viewing behaviors. By employing NinjaHeads, users can effectively perform operational and navigational tasks while gaining access to blind spots in occluded areas, as if they have more than just a pair of eyes. Moreover, with 3D reconstruction technology, it can find applications in industrial scenarios, such as crane or forklift operations, to enhance depth perception precision and user safety, thus preventing accidents.

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